Fundamentals of Software Project Management (SE4002)

Date: Nov 5th 2025

Course Instructor(s)

Mr.Minhal Raza

Re Mid-2 Exam

Total Time(Hrs): 01

Total Marks: 28 Points

Total Questions: 04

Roll No ____________ Section ____________ Student Signature ____________

Do not write below this line

Attempt all the questions. (NPV-Table is Allowed)

CLO #2: Critically evaluate and discuss the issues around project management and its application in the real world with course participants and learners

Q1 : Read and comprehend the following case study and answer the questions at the end of the case study **[8-Points, Est. Time : 15 Mints]**

FASTEd Pvt. Ltd., an educational technology company, plans to expand its online learning platform **"TechBridge"** to reach rural schools across Pakistan. The project aims to install digital learning kiosks, train local teachers, and ensure uninterrupted access to internet-based lessons. During the initial planning, the project manager, Sara, realizes that her internal IT team lacks experience in hardware installation and rural logistics. Some team members express concern that hiring external vendors could cause budget and coordination problems. Meanwhile, a few local community leaders are showing hesitation about the company's motives, demanding employment for local people instead of bringing staff from cities. In the mid-planning phase, a severe fluctuation in global hardware prices makes it unclear whether the company should lock a vendor contract early or wait for potential price drops. Sara also notices that her team morale is dropping because of unclear role assignments and frequent task changes caused by uncertain decisions. Unexpectedly, one vendor offers an urgent limited-time proposal to supply low-cost equipment, but it comes with strict terms and penalties if the delivery schedule is not met. Sara must decide quickly—without full approval from some key stakeholders who are currently unavailable. **Identify** which **knowledge areas** (*Multiple areas overlap — think carefully about hidden connections*) are involved in this case, and explain how the project manager should apply them to make **a balanced and effective plan**.

Answer : Project Integration Management – Sara needs to balance all aspects (scope, cost, time, and stakeholder needs) and make an integrated decision that aligns with project goals. Stakeholder Management – She must address community concerns, involve local leaders, and communicate transparently to gain trust and support. Procurement Management – Vendor selection, contract timing, and terms require careful analysis to avoid future risks and penalties.Human Resource (Resource) Management – Low team morale and unclear roles need attention through proper role definition and motivation. Risk Management – Fluctuating prices and vendor risks must be assessed, and contingency plans should be made. Communication Management – Sara should ensure clear, regular updates to both internal teams and external stakeholders to reduce confusion. Cost Management – Careful budgeting and cost forecasting are needed before locking any vendor contracts.

How to Apply: Sara should integrate all plans, communicate clearly with her team and stakeholders, evaluate vendor risks and costs before contracting, define team roles to improve morale, and involve local communities to build trust. This balanced approach will ensure both project success and stakeholder satisfaction.

Q2 :Read and comprehend the following case study and answer the questions at the end of the case study **[6-Points, Est. Time : 10 Mints]**

Project Helix was launched by a global health analytics firm to create a real-time patient monitoring system that integrates wearable sensors, cloud analytics and hospital dashboards. The initiative was led by Ayla, a technically brilliant data architect recently promoted to project lead. Though confident in her technical judgment, she preferred minimal managerial structure to “keep innovation flowing.”
To speed delivery, Ayla encouraged each subteam — software, data, and device integration — to work independently and “synchronize later.” Documentation was replaced by brief chats and shared code snippets. Early prototypes impressed executives, but as modules grew, data inconsistencies and latency spikes appeared. The analytics engine required database schema changes, which the device team wasn’t informed about, leading to silent system crashes. Tensions rose when the cloud vendor introduced a new pricing model, increasing project costs, but no one knew who was responsible for vendor coordination. Ayla suspected poor coding and ordered a full refactor, but deadlines loomed, and developers reused half-tested modules to save time. When the system was finally demonstrated, sensor readings fluctuated wildly, reports were delayed, and dashboards froze under real-time load.
Management blamed both the team and the technology, but no clear failure point could be identified. The project was paused for an audit to determine whether leadership, process design, product architecture, or technology decisions had caused the breakdown. **Identify and justify** which **classical mistakes** occurred in the areas of **People, Process, Product, and Technology.** Explain how these issues are interconnected, and propose specific corrective actions that could have prevented Project Helix’s collapse.
Answer: People: Weak leadership and poor coordination — Ayla focused only on technical work, not team management. Lack of clear roles and accountability — no one managed vendor or communication tasks. Process: No structured communication or documentation. Poor integration planning — teams worked in isolation and “synchronized later.” Product:Uncontrolled changes in database and code caused mismatched modules and crashes. Incomplete testing and rushed reuse of half-tested code.
Technology: Over-reliance on untested integrations and rapidly changing cloud services without stability checks. Interconnection: Poor leadership (People) led to weak processes (Process), causing mismatched product components (Product) and unstable system performance (Technology).
Corrective Actions: Define clear roles, responsibilities, and communication channels. Maintain proper documentation and version control. Plan early integration and testing across all modules. Assign a vendor coordinator and monitor technology changes. Regular progress reviews and risk assessments.

CLO #1: Explain principles of the project life-cycle and how to identify opportunities to work with learners on relevant and appropriate project scenarios to share this understanding

Q3 : Below is cash flow table for 3 different projects (A, B, C) over 6 years, showing positive and multiple negative values (in Rs, 000) a software house considering for investments: **[8-Points, Est. Time : 15 Mints]**

Year	Project A	Project B	Project C
0	−400	−500	−300
1	120	180	−100
2	140	−150	130
3	−80	220	160
4	200	250	−70
5	260	−100	220
6	180	310	260

Additional Information :
Discount Rate (r) = 11% per year
Residual/Salvage Value at the end of Year 6: Project A: 50k,Project B: 70k,Project C: 30k and Maintenance Cost each year after Year 1 for project: A = $20k, B = $25k, C = $15k
Corporate Tax Rate = 20% on positive cash inflows after accounting for maintenance.

I. Calculate the final cash flows and net profit.

Reject Project “B” has a negative net profit so we don't need to calculate all for Project B.

II. For each project comment and Rank by using cash-flow forecasting method.

“Project A”, “Project C” , Always Reject Project “B” .

III. Preform Cost Benefit Analysis for each project and comment which project is more beneficial?

“Project A”, “Project C” , Always Reject Project “B”.

IV. Calculate ROI ,NPV for each project and comment which project is more suitable?

“Project A”, “Project C” , Always Reject Project “B”.

V. Calculate IRR under window of 10% and 20% of the interest rate.

VI. Calculate Payback and Discounted Payback for each project , compare and comment your results.

“Project A”, “Project C” , Always Reject Project “B”.

VII. What if I change the rate from 11% to 20% . Explain theoretically.

“Doesn't Affect Any Thing As NVP Is Still in Minus”

[Only share 1st Calulation , Relative Marking for Question , If Calculation for Project A is Correct”]

Project A	Cash Flows	Maintaince	After Maintaince	After Tax	Total Cash Flow	NPV (11%)	NPV	PayBack	PayBack Discounted
0	-400,000				-400,000	1	-400000	-400000	-400000
1	120,000	0	120,000	120,000*0.8	96,000	0.901	86496	-304000	-313504
2	140,000	20,000	120,000	120,000*0.8	96,000	0.812	77952	-208000	-235552
3	-80,000	20,000	0	0	-100,000	0.731	-73100	-308000	-308652
4	200,000	20,000	180,000	180,000*0.8	144,000	0.659	94896	-164000	-213756
5	260,000	20,000	240,000	240,000*0.8	192,000	0.593	113856	28000	-99900
6	180000 +50,000 =230,000	20,000	210,000	210,000*0.8	168,000	0.535	89880	196000	-10020
	Net Profit				**196,000**		**-99900**		
ROI = Avg.Net Profit / Cost of Investment		Payback = b/w 4 and 5th year							
Avg Net Profit = (196,000)/5 =	32666.66667	Payback=4+(164,000/192,000)							
ROI =32666.67/400000 =	0.081666675	8.17%	4.85 years		No Discounted Payback Period				

Question-4 :Read and comprehend the following case study and answer the questions at the end of the case study **[6-Points, Est. Time : 15 Mints]**

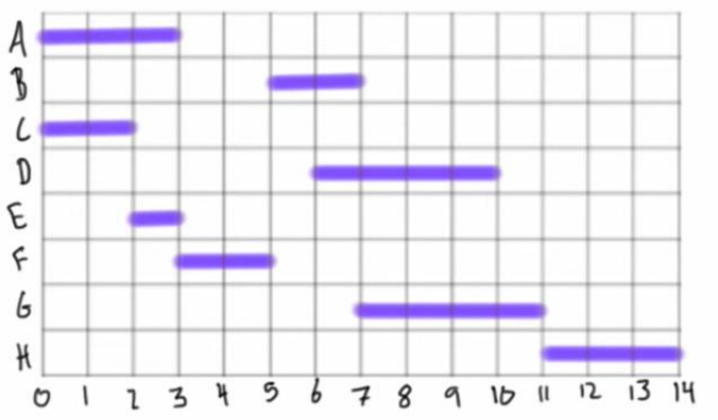
The company TechNova Solutions has taken on a project to develop a web-based project tracking system called TrackEase for a medium-sized enterprise. The project manager has broken down the development process into several activities, each with specific dependencies and timing relationships.

The project begins with Activity A (Requirements Analysis), which takes 3 days to complete. Once the analysis is mostly done, Activity B which have 2 days duration (User Interface Design) can start, but only 2 days after the completion of the analysis. At the same time, Activity C (Database Design) can start simultaneously with the start of requirements analysis and is expected to take 2 days. After the interface design begins, Activity D (Frontend Development) can start 1 day after the start of Activity B and will take 4 days to complete. Meanwhile, Activity E (API Integration Planning) depends on Activity C but is scheduled to start 3 days before C finishes, lasting 1 day. Activity F (Backend Development) is also dependent on Activity C, and it must finish 3 days after C finishes — with a total duration of 2 days. When both D and E are ready, Activity G (System Integration) begins; it starts 1 day after D starts and runs for 4 days. Finally, Activity H (Testing & Debugging) begins only 2 days before F finishes and after G is complete, taking 3 days to finish.

The project manager must carefully analyze the sequence and inter dependencies to create a proper schedule and identify the duration of the TrackEase project. Help him by using “2” time management appropriate Tools.

Answer:

Activity	Predecessor	Duration
A	-	3
B	A(FS2)	2
C	A(SS)	2
D	B(SS1)	4
E	C(SF3)	1
F	C(FF3)	2
G	D(SS1), E	4
H	F(SF2), G	3



“OR”

TechNova , a software development company is considering designing a cloud-based inventory management system for the retail stores and is currently engaged in a cost-benefit analysis . Market research shows that , if TechaNova launches early and efficiently , with no strong competitors . It could achieve a high sales level generating an annual income of Rs.900,000 . It estimates that there is a 15% chance of this happening.

However, a larger rival might release a similar system before TechNova’s launch date, which could reduce its annual income to only Rs,150,000. It estimates that there is a 25% chances of this occurring. The most likely outcome, according to its analysis , is that TechNova will launch around the same time as its competitors and well achieve a moderate income of Rs,600,000 with a 60 % probability. TechNova h as therefore calculated its expected income as shown in Table 1.0 below:

Sales Level	Annual Sales Income (Rs.)	Probability (p)	Expected Value (E =i*p)
High	900,000	0.15	135,000
Medium	600,000	0.60	360,000
Low	150,000	0.25	37,500
Expected Income			532,000

Development costs are estimated at Rs.800,000.Annual marketing and maintenance costs are projected to be Rs.250,000 regardless of market performance.Sales are expected to remain steady for at least 4 years. Would you like to advise TechNova to proceed with the project on the outcome.Support your answers with appropriate calculations and reasoning.

Answer:

Given Data

Item	Value (Rs.)
Development Cost (one-time)	800,000
Annual Marketing & Maintenance Cost	250,000 per year
Expected Annual Income	532,000
Project Life	4 years

Annual Net Income

Net Annual Income=Expected Income−Annual Cost =532,000−250,000=282,000=

Total Net Income Over 4 Years

282,000×4=1,128,000

Net Profit (After Recovering Development Cost)

Net Profit=1,128,000−800,000=328,000

Decision : The project gives a positive net profit of Rs. 328,000 over 4 years Even though risks exist (competition and timing), the expected return is higher than total costs. The moderate income scenario (Rs. 600,000 with 60% probability) is strong enough to cover expenses and ensure profit. TechNova should proceed with the project. The expected return is favorable, risks are manageable, and long-term benefits outweigh the initial investment. The project's expected income (Rs. 532,000 per year) comfortably covers yearly costs and recovers the initial investment within the project duration, making it financially viable.